Improving the Campaign Retrieval Model for High-Level Waste Vitrification Through LNS

A large-neighborhood search over retrieval campaigns, its architecture, and what it can and cannot save

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Rev. 0

Executive summary

This paper documents a new optimization layer for the Hanford high-level waste (HLW) campaign retrieval model. The retrieval model, which follows Section 6.2 of RPP-RPT-64878, walks the tank farms in a fixed order and assembles blends of up to five tanks per retrieval campaign inside a 450 to 550 kgal volume band. The new layer improves on that baseline with a large-neighborhood search (LNS): it keeps the baseline campaign set, repeatedly destroys small groups of campaigns, rebuilds them under all of the same sequencing and waste acceptance rules, and keeps a rebuilt set only when the exact glass formulation model confirms that the mission needs less glass. Because acceptance is strictly improving, the method cannot return a mission worse than the baseline it starts from.

The work began as an attempt to reduce the number of HLW canisters produced over the mission. The central result is a negative one, and it matters: a careful statistical study of the exact glass formulation model showed that inverse waste loading is nearly linear in blend composition, which makes total mission glass almost independent of how waste is partitioned into campaigns. At a fundamental level this result is not surprising given that the inventory of High-Level Waste in the source tanks is unchanging across the whole mission. The real optimization exists in the ability to segregate the High-Level portion and the Low-Level Waste (LLW) streams to an alternative location. Given that this is outside the scope of this paper, the focus here will be on what a LNS algorithm does bring to the existing paradigm of campaign formulation.

Multiple independent campaign builders, each smarter than the baseline in some measurable way, all produced more canisters than the baseline they replaced (+2,092 and +708 canisters on the reference scenario). The canister count is effectively at a floor. What the baseline does leave on the table is campaign count: the paper's own builder emits many tiny sub-band campaigns where its conservative screening limits reject every in-band blend.

Across all 32 published mission scenarios, the LNS reduced the total campaign count across formulations from 2,284 to 1,827 (a 20.0% reduction) while reducing projected glass by 667 tonnes (0.057%), corresponding to 216 fewer canisters. Seventeen scenarios improved on canisters, fifteen were unchanged, and none got worse. Every result passed the full hard-rule verification battery: Only blending is practical within a local quadrant, the five-tank limit, layer stack order with in-flight carryover, southeast tank precedence, the volume band, minimum transfer volumes, exact mass conservation, and waste acceptance compliance. On the reference scenario 135.120.150, the search collapsed a 16-campaign chain of capped campaigns into 5: one

in-band cross-layer blend, one in-band pure run, and a legal final drain, with two baseline pairings surviving, by cross-blending layers of the same tank that the baseline's screening limits had kept apart.

Two methodological findings generalize beyond this application. First, when a set-value function is nearly linear in the composition of each set, greedy repartitioning has nothing to win and variance to lose; improvement searches that start from a feasible incumbent and accept strict improvements only are the correct tool. Second, blending is not always good: individually hot layers can blend to a worse waste loading than they achieve alone, so a repair step should generate both blended and de-mixed candidates and let exact evaluation arbitrate. Both findings are quantified in the Results and Discussion sections.

Introduction

The retrieval planning problem

The HLW mission moves waste from 177 underground tanks to a vitrification plant. Waste is retrieved tank by tank and layer by layer, blended into campaigns of up to five tanks each, and fed to a joule-heated melter where it is immobilized in borosilicate glass. The number of canisters produced and the total time to immobilize is the mission's dominant cost driver. The number of canisters is set by two quantities per campaign: the mass of waste oxide fed, and the waste loading (WL) the glass formulation model can achieve for that particular blend. Total canisters follow from the sum of glass over all campaigns, at 3,097 kg of glass per canister (the Appendix D planning value).

RPP-RPT-64878 Section 6.2 defines the baseline construction, but only for the SE quadrant. This serves as a starting point for extrapolating the mission to the full farm. Rules set are as follows: inside a quadrant, respect tank precedence rules (AP-101 before AP-102, AZ-101, AZ-102, and AY-101, and similar); keep an in-flight layer running until it is empty; never tap a layer above an undrained one; keep every campaign between 450 and 550 kgal of slurry; take at least 38 kL from any layer touched, unless the take empties the layer; and blend no more than five tanks. Within each campaign a linear program maximizes waste mass subject to linearized waste acceptance criteria (WAC) limits, which the paper scales to 60% of the regulatory values (90% for fissile ratio and NOx) as a mixing allowance. The LP caps each layer's take at one fifth of the band maximum.

This baseline is a sequential greedy walk. It never revisits a campaign once built, and it applies its screening limits conservatively at every step. The question this work set out to answer is whether a global optimizer, given the same hard rules and an exact evaluation of every candidate mission, can do better: fewer canisters, fewer campaigns, or both.

Why the problem is hard

Three properties of the problem shape everything that follows. First, exact evaluation is expensive. Determining the waste loading of a candidate blend requires solving the EWG2.5 glass formulation problem (PNNL-35884): a multi-start sequential quadratic programming solve over product consistency, viscosity, electrical conductivity, sulfate solubility, phosphate capacity, and composition validity constraints, taking about 1.5 seconds per blend on a desktop

workstation. A mission has tens of campaigns; a search that evaluates even a few thousand candidate blends runs for hours.

Second, the decision space is combinatorial. The southeast quadrant alone contributes on the order of a hundred thousand possible five-tank groupings, each with a continuous allocation of volumes inside it. Any practical method must decouple candidate generation (cheap, linear) from exact evaluation (expensive, nonlinear).

Third, and most consequentially, the objective is nearly insensitive to the partition. The Results section on the linearity result quantifies this: fitted on 76 exact campaign evaluations, a linear model of inverse waste loading on oxide mass fractions achieves $R^2 = 0.79$. Because total mission glass is the waste-weighted sum of inverse waste loading, and because mass conservation fixes the waste-weighted total of each oxide across any partition, the linear part of the objective is identical for every legal partition. What remains is a residual with a standard deviation of about 0.5 on a mean inverse waste loading of 2.6, poorly predicted by quadratic corrections (20% of the residual variance). The residual is real, but only exact evaluation resolves it.

What was built

The optimizer lives in a new package, optimized_mission_DRAFT, alongside the existing campaign builder. It contains a re-implementation of the Section 6.2 baseline used as the starting incumbent, an exact evaluation path shared with the production glass formulation code, a blend-signature memoization cache, three classes of ruin operators, two classes of repair (blended and de-mixed), and a verification-gated acceptance rule. A sweep driver runs the whole pipeline over all 32 published scenarios with parallel workers. Every number in this paper comes from these programs run against the cached October 2026 Best-Basis Inventory dataset, and every headline figure was re-derived from the cached reports during writing.

Scope and reading guide

Results are presented first, followed by Methodology, which documents the architecture and algorithms in the depth an engineer needs to reproduce a number or audit a rule. Discussion covers the interpretation of the linearity result, the cost of retrieval-legal acceptance, limitations, and data provenance. Conclusions state what should change in practice. All quantitative results were produced by the code on the optimized-mission branch of the repository as of 13 October 2026.

Results

The linearity result, and why it reframes the problem

The study began with three from-scratch campaign builders, each of which ranked candidate tank groupings by a learned or heuristic prediction of blend quality and built the mission greedily. All three lost to the baseline they were meant to beat. The surrogate-ranked builder gave +2,092 canisters on the reference scenario; a calibration of the ranking model to exact evaluations still gave +708. Before building a fourth variant, we asked why every variant failed in the same direction.

The answer is in Figure 1. Fitting inverse waste loading linearly on the twelve most prevalent oxide mass fractions reproduces the exact EWG2.5 evaluations of the baseline's 76 campaigns with $R^2 = 0.79$. Total mission glass is the waste-weighted sum of inverse waste loading over campaigns. Write the fit as 1/WL = b_0 plus a linear term in the oxide fractions plus r(f), where r is the residual. The glass sum then separates into three parts: a constant times total waste oxide mass, the oxide-slope terms times the mission-total mass of each oxide, and the residual part, the campaign sum of waste mass times r(f). Mass conservation pins the first two parts regardless of how waste is partitioned into campaigns. Only the residual term depends on the partition at all.

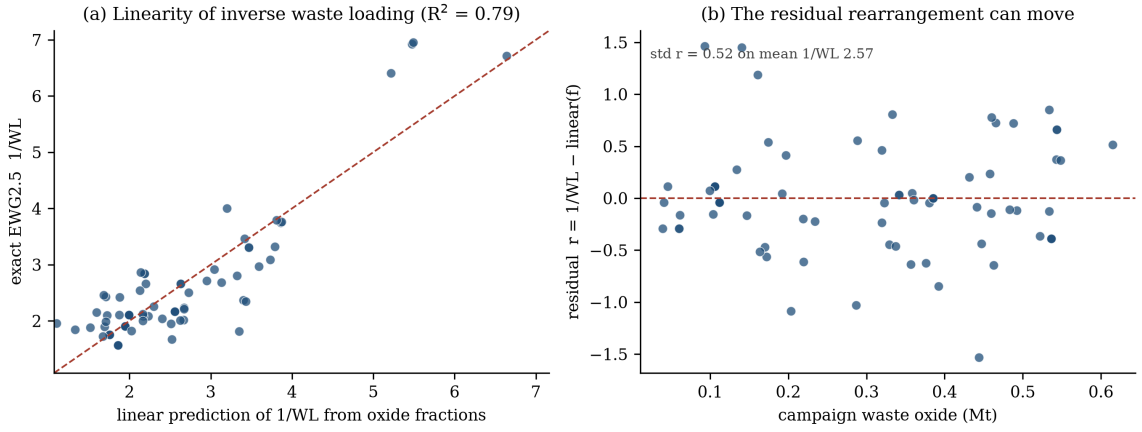


Figure 1. The linearity result on scenario 135.120.150. (a) Exact EWG2.5 inverse waste loading versus the linear prediction from oxide mass fractions across the baseline's 76 campaigns ($R^2 = 0.79$). (b) The residual, the only partition-dependent term, plotted against campaign waste mass; it has standard deviation 0.52 on a mean 1/WL of 2.57 and is poorly predicted by quadratic corrections ($R^2 = 0.20$).

Two consequences follow. First, any greedy builder that ranks candidates by predicted quality is optimizing an objective that is constant across its choices; the variance it introduces in the residual term (through unlucky blends that only exact evaluation would catch) is pure loss, which

is exactly the pattern observed. Second, the theoretical ceiling on composition-driven improvement is set by the residual, and the residual is small: an optimistic reading of Figure 1(b) suggests that shifting every campaign to the favorable side of its residual could save on the order of one percent of mission glass. The observed suite-wide saving, 0.057%, sits comfortably under that bound.

A complementary bound comes from campaign packing. The baseline's sub-band campaigns carry waste at a lower effective waste loading (0.383 blended over all sub-band campaigns on the reference scenario) than in-band campaigns (0.420). Merging all sub-band waste into in-band campaigns at in-band quality would save 694 canisters, 3.9% of the mission, if waste loading were the only consideration. This is an upper bound, not a promise: some sub-band campaigns are pure runs of layers whose waste loading exceeds the in-band average, and merging them into blends lowers it. The point is that the structural headroom is an order of magnitude larger than the compositional headroom, so the search concentrates on structure.

Suite results across all 32 published scenarios

Table 1 summarizes the sweep over all 32 published mission scenarios. The LNS ran 60 iterations per scenario with a 900-second wall budget and 10 parallel scenario workers; total compute was 7.6 core-hours. Seventeen scenarios improved on canisters, fifteen were unchanged, none worsened. Campaign count fell on 28 scenarios, was unchanged on 3, and rose on 1 (a deliberate canister trade discussed below); the suite total dropped by 457 campaigns, 20.0%.

Table 1. Suite results over the 32 published scenarios (baseline §6.2 builder versus LNS, exact EWG2.5 evaluation).

Metric	Baseline	LNS	Change
Total campaigns	2,284	1,827	-457 (-20.0%)
Total glass (t)	1,169,447	1,168,780	-667 (-0.057%)
Total canisters	377,607	377,391	-216 (-0.057%)
Scenarios better / equal / worse (canisters)	—	17 / 15 / 0	—
Scenarios passing verification	32 / 32	32 / 32	—
Exact evaluations (cache hits / misses)	—	116,895 / 3,970	29.4 : 1
Compute (core-hours, whole sweep)	—	7.6	—

Figure 2 shows the per-scenario detail. Campaign reductions cluster between 5 and 32 per scenario; the largest absolute cuts are on the low-solids scenarios (135.120.50 loses 32 of 125 campaigns, 135.105.50 loses 31 of 119), which have the longest chains of capped campaigns to repack. Canister savings concentrate on five scenarios (135.120.100 at 54, 135.120.200 at 51, 130.120.100 at 27, 135.115.50 at 16, 125.120.100 at 15); the median scenario improves by less than one canister.

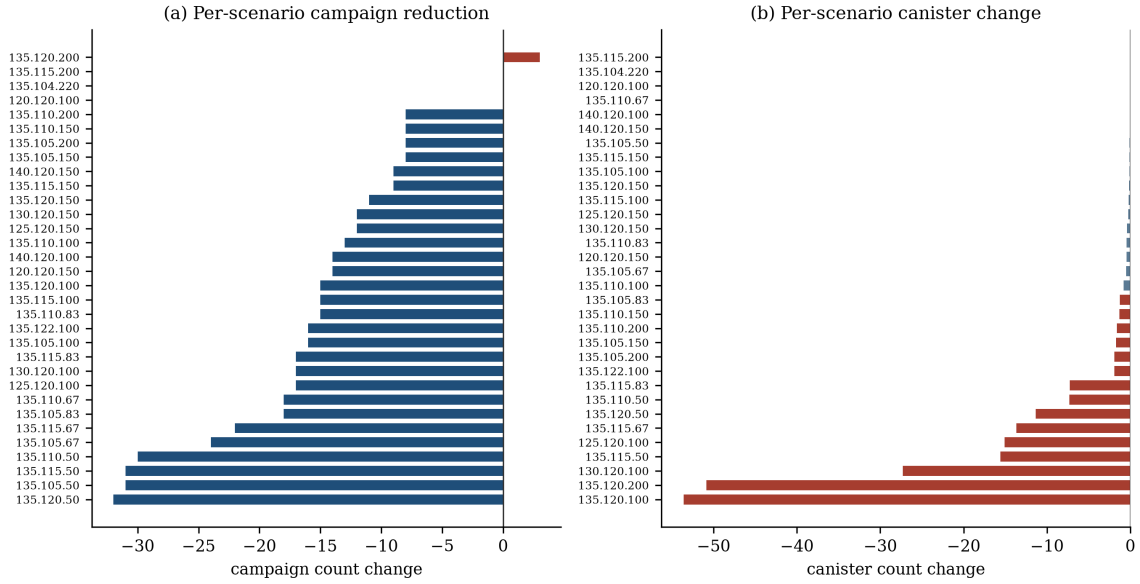


Figure 2. Per-scenario changes across the 32-scenario sweep. (a) Campaign count change; one scenario (135.120.200) gains 3 campaigns, trading structure for 51 canisters. (b) Canister count change; no scenario worsens.

One scenario gained campaigns: 135.120.200 went from 50 to 53 campaigns while saving 51 canisters. This is the correct trade under the objective (canisters dominate), but it deserves mention because campaign count, not canisters, drives much of the operational workload: each campaign carries its own sampling, WAC evaluation paperwork, glass-former recipe, and heel management. Where a team values campaign count explicitly, the acceptance rule can weight it; the shipped objective does not.

The reference scenario in detail: campaign structure

Scenario 135.120.150 is the paper's own validation condition and the most instructive case. The baseline emits 76 campaigns, of which 48 are below the band floor. Twenty-eight of those are a single artifact: tank AN-106 holds three stacked layers that individually violate the 60%-scaled slurry-basis screening limits on glass-target species (GTar_NaK runs 0.99 to 1.11 times the

scaled limit on the layer values; GTar_P up to 1.77; GTar_CrSol 1.42). No in-band blend containing much of any AN-106 layer survives screening, so the builder's fallback emits a chain of 16 pure-layer campaigns capped at 416 kL each, one fifth of the band maximum, followed by residual heel campaigns.

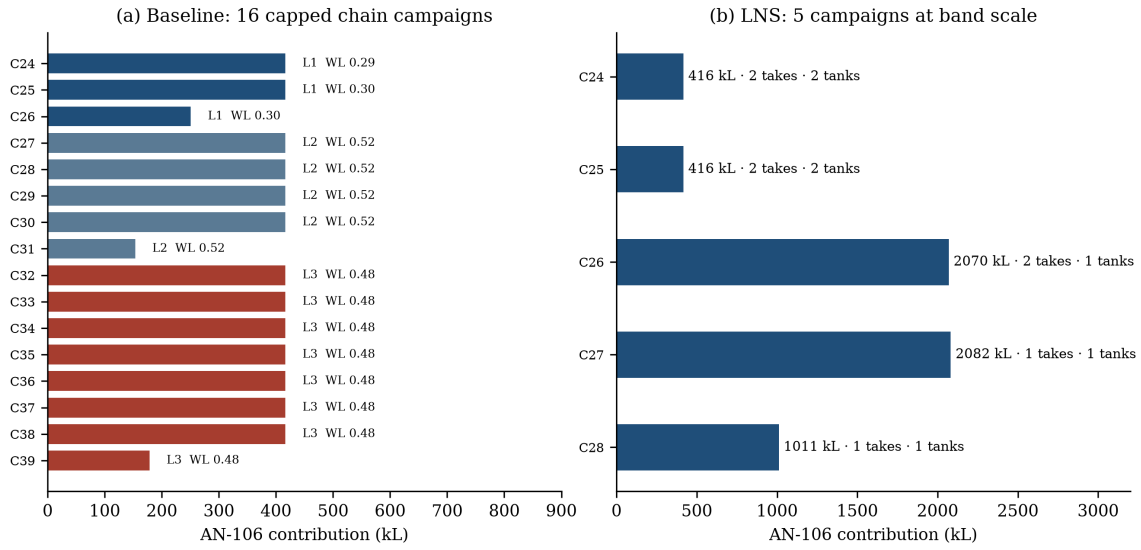


Figure 3. The AN-106 chain on scenario 135.120.150. (a) The baseline's 16 capped pure-layer campaigns (colors distinguish layers L1, L2, L3; labels show layer and exact waste loading). (b) The LNS repack: the two baseline blend campaigns that remain unchanged, one in-band cross-layer blend (L1 with L2), one in-band pure L3 run, and the legal final drain; per-layer caps removed.

The chain repack is the search's clearest structural win. Destroying the whole chain at once (the tank-chain ruin operator) frees all three layers simultaneously, and the repair LP, with per-layer caps removed and screening limits climbed to their true values, produces an in-band cross-layer blend (2,070 kL of L1 with L2 at waste loading 0.485) and an in-band pure run of L3 (2,082 kL at 0.475), replacing eight capped campaigns with two in-band ones. Exact evaluation confirms both pass, and the mission's campaign count drops from 76 to 65 in one accepted move. The move was found in iteration 1 of 60; the remaining 59 iterations found nothing better, which is itself informative: once the chain artifact is repaired, the incumbent is at the residual ceiling.

The same case demonstrates why the acceptance rule must gate on exact evaluation, and why blending must compete against not blending. The two baseline blend campaigns that pair AN-106_L1 with AP-102_L3 (416 kL each, waste loading 0.289 and 0.302) survive the repack untouched because the arbitration step found no better use for that pairing; pure AN-106_L1 alone evaluates at 0.302, no better than the blend it sits inside. Meanwhile the new in-band L1/L2 blend runs at 0.485 against 0.524 for pure L2 and 0.302 for pure L1: the blend is worse than

the better pure layer. The classical intuition that blending always raises waste loading is simply wrong when the blend partners are hot on the same glass-limiting species. Total glass moved by less than 2 parts in 10,000 on this scenario; the -1 canister and the 11-campaign reduction both come from structure, not composition.

Verification evidence

Every accepted incumbent, and therefore every published result, passed the nine-part verification battery: quadrant purity; at most five tanks per campaign; layer stack order under mission-global in-flight semantics (a deeper layer may only start in the campaign where the layer above drains); southeast tank precedence; the 450 to 550 kgal band with sub-band campaigns permitted only as final drains; the 38 kL minimum transfer with final-drain exemption; exact mass conservation on every layer; waste-oxide conservation on the mission; and WAC evaluation of every campaign blend against the true regulatory limits. Mass conservation closes to floating-point exactness (residuals below 1e-12 of layer volume).

The sequencing gate is not decorative. An earlier development sweep without it accepted moves whose repair splices started a deeper layer before the one above drained, or started a successor tank before its predecessor finished. That sweep measured 849 canisters of suite-wide savings, and 15 of 32 scenarios failed verification. Figure 4 shows the comparison. The shipped gate rejects such moves outright; the baseline starts legal, so the incumbent stays legal by construction. The 633-canister difference between the sweeps is the honest price of retrieval-legal optimization, and reporting it guards against someone later re-introducing the bug in pursuit of the larger number.

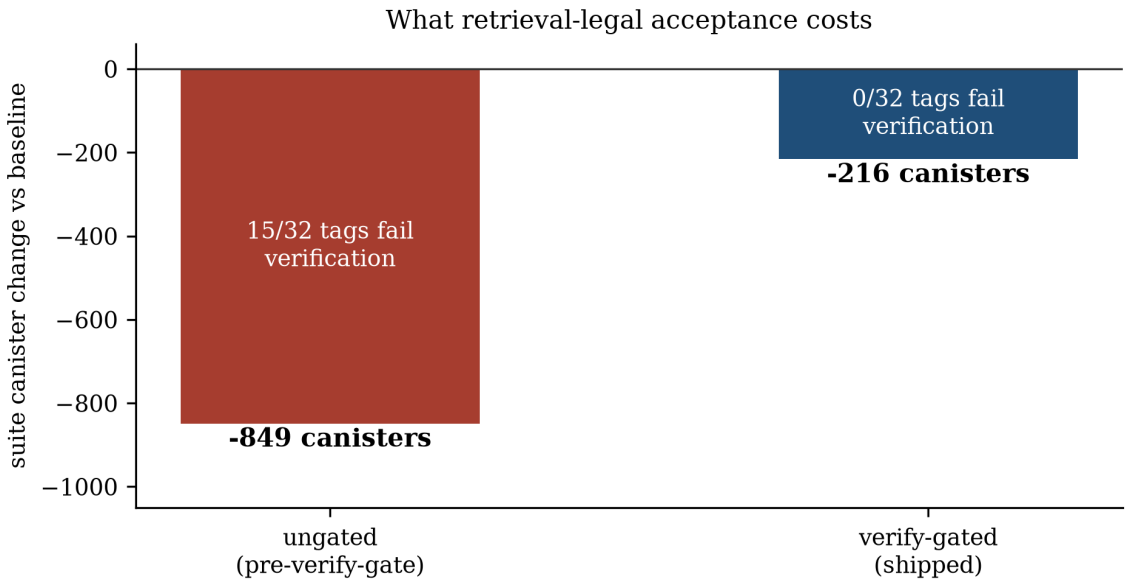


Figure 4. Suite-wide canister savings with and without the retrieval-sequencing verification gate. The ungated search finds 849 canisters but breaks layer-stack and tank-precedence rules on 15 of 32 scenarios; the gated search ships 216 canisters with all scenarios legal.

Computational performance

The blend-signature memo is what makes the search affordable. Every exact evaluation is keyed by the rounded per-layer take vector of its campaign, so re-encountering an identical blend is free. Across the suite the cache served 116,895 hits against 3,970 misses, a ratio of 29.4 to 1; on the reference scenario the ratio was 31 to 1, saving roughly two hours of sequential quadratic programming per run. Per-scenario wall times ranged from 11 to 24 minutes (median 14.5) on a 10-core desktop, with scenario-level parallelism across workers. The baseline rebuild takes about 10 seconds per scenario.

Methodology

Architecture overview

The optimizer is a Python package of five modules with a deliberate dependency direction: everything points inward toward the exact evaluation path, which is shared with the production glass formulation code so that a number accepted by the search and a number reported by the app can never disagree. The objective module owns the problem data (materials per layer, oxide vectors, WAC values, the exact evaluation wrapper, and the mission rollup). The campaign_optimizer module owns the rule layer: the per-campaign linear program, the Section 6.2 sequencing rules, and the scaled-limit ladder. The lns module owns the search. The verify module owns the hard-rule battery. The sweep and aggregate modules own multi-scenario orchestration and reporting. Figure 5 shows the search loop and the data flow through its gates.

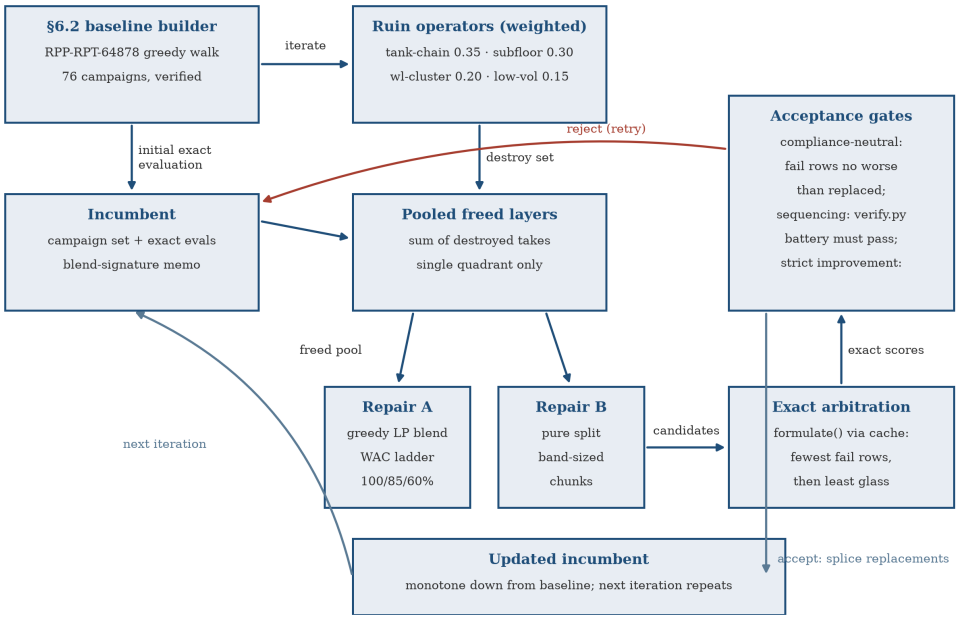


Figure 5. The LNS loop. The incumbent starts as the verified §6.2 baseline. Each iteration selects campaigns through a weighted ruin operator, pools their freed layer volumes, generates two repair candidates (blended and pure-split), scores both with exact glass evaluation through the memo cache, and splices the winner into the incumbent only if it passes the compliance and sequencing gates with strictly less glass.

Problem statement and conventions

A campaign is a partial assignment of layer volumes: for each layer it touches, a take in kilolitres. A mission is an ordered list of campaigns. Hard rules, all verified post hoc: campaigns

draw from one quadrant; at most five tanks per campaign; layer stacks drain top-down with in-flight carryover only at the boundary campaign; southeast tank precedence; campaign volume in [1,703.4, 2,082.0] kL except a legal final drain; takes of at least 38 kL or complete drains; every layer fully consumed by the mission's end. The objective is total glass divided by 3,097 kg per canister, rounded up, with each campaign's glass equal to its waste oxide mass divided by its waste loading, where the waste loading comes from the EWG2.5 formulation solve.

Waste acceptance criteria enter at two levels that must not be confused. Inside the repair LP, WAC rows are the paper's slurry-basis linearization at the §6.2 scaled limits; these rows are a fast, conservative screening approximation. At acceptance, compliance is judged on exact glass-level evaluation, which is the regulatory basis and the same basis the app's reports use. A blend the screening rows reject can pass exactly, and one that screens clean can fail exactly; only the exact result decides.

The exact evaluation path and the blend-signature memo

Exact evaluation of a campaign takes its layer takes, computes the blended oxide vector by volume-weighting the layer vectors, and solves the EWG2.5 formulation (multi-start SLSQP over the property constraints of PNNL-35884) for the maximum waste loading. The wrapper returns success, waste loading, and a structured evaluation object whose per-criterion rows carry the compliance verdicts used by the acceptance gate.

Because the search re-encounters blends constantly (a rejected move's campaigns return on the next iteration; a repaired campaign often reproduces an existing one with a different take split), every evaluation is cached under a signature computed from the sorted, rounded per-layer take vector. Rounding at 1e-6 kL makes the key robust to linear-program noise while keeping genuinely different blends distinct. Cache hits cost a dictionary lookup; misses cost 1.5 seconds of SLSQP. The measured 29.4-to-1 hit ratio across the suite means the search bought 29 expensive evaluations for the price of one.

Ruin operators

Each iteration picks one of four operators by fixed weights and destroys the campaigns it selects. The operators are chosen to match the failure modes observed in the baseline, and their weights were tuned on the reference scenario before the sweep.

The tank-chain operator (weight 0.35) selects every single-tank sub-band campaign of the tank with the most such campaigns. It exists because the baseline's capped pure-layer chains cannot

be repacked piecemeal: repairing one campaign of the 16-campaign AN-106 chain just produces another capped campaign, since the per-layer cap and the band floor make any single-chain-campaign pool infeasible. The whole chain must be freed at once. This operator found the reference scenario's only accepted move.

The subfloor-group operator (weight 0.30) selects the k worst-waste-loading sub-band campaigns of one quadrant, grouping their pools for a joint repair. The worst-WL-cluster operator (weight 0.20) takes the worst sub-band campaign plus its nearest neighbors in oxide-composition space, putting candidate dilution partners in the same repair pool. The low-volume operator (weight 0.15) takes the k smallest campaigns of a random quadrant as a pure structure move. Destroy sizes of 4 to 6 campaigns are typical; the sweep used 6.

Dual-candidate repair with exact arbitration

Repair rebuilds the freed pool into new campaigns. Two structurally different candidates are always generated, and the choice between them is made by exact evaluation, not by a heuristic.

Repair A is a greedy blended repair. It repeatedly solves, over all tank tuples of size up to five drawn from the freed pool, the campaign LP that maximizes oxide transit subject to the volume band and the WAC screening rows, commits the best solution's takes, and repeats until the pool is exhausted. Two changes relative to the baseline builder matter: the per-layer take cap is removed (the cap is what creates chains), and the WAC screening rows are climbed on a ladder from the true regulatory limits down through 85% and 60% scalings, accepting the first level that admits any in-band solution. The ladder direction deserves emphasis: climbing to true limits first is not a compliance relaxation, because the exact glass-level evaluation at the acceptance gate still decides, and the gate is compliance-neutral.

Repair B is a pure split: every freed layer runs alone, in band-sized chunks, ordered by tank precedence and stack index. Pure runs avoid the possibility that blending lowers waste loading, which the AN-106 case showed is real: the baseline blends AN-106_L1 with AP-102_L3 at waste loading 0.289 to 0.302 while the pure layers run 0.30 to 0.52. The classical intuition that blending always helps dilution fails when the blend partners are individually hot on the same glass-limiting species.

Arbitration scores both candidates on three keys in order: fewest failed regulatory rows on exact evaluation, then least exact glass, then fewer campaigns. The winner becomes the move's proposed replacement set. On the reference scenario, arbitration selected the blended repair for

the chain move after the pure split scored worse; on other moves it has selected the pure split. Letting the expensive-but-honest evaluation decide is the point of the memo cache.

Acceptance: three gates

A proposed replacement set is spliced into the incumbent at the destroyed campaigns' positions and then must pass three gates.

The compliance gate is neutral rather than strict: the worst exact-evaluation failure count among the rebuilt campaigns may not exceed the worst failure count among the campaigns they replace. A strict gate (every rebuilt campaign must pass every row) rejects all progress on the northwest and southwest heel campaigns, whose NOx exceedance is a property of the heel material itself; the baseline carries those failures, and no blend can remove them. The neutral gate guarantees the move never degrades compliance relative to the status quo it replaces, which is the strongest guarantee available without inventing dilution that the data does not support.

The sequencing gate runs the full hard-rule verification battery on the spliced mission and rejects the move on any violation. This gate was added after the ungated development sweep produced retrieval-illegal missions on 15 of 32 scenarios; the failure mode is structural, because a repair LP has no notion of campaign order, so splicing repaired campaigns at the destroyed block's position can start a deeper layer early or jump a tank precedence. Since the initial incumbent (the baseline) is verified legal, gating acceptance keeps every subsequent incumbent legal by construction, and the final published result is legal by the same argument.

The improvement gate is strict: the spliced mission's exact total glass must be lower than the incumbent's. Together with the deterministic evaluation path, this makes the search a hill climb with a monotone descending objective: the sequence of incumbent glass values cannot increase, so the final result is never worse than the baseline regardless of the random seed. The flip side is that the search can stall at a local floor, which the operator weights and destroy sizes are tuned to mitigate. On the reference scenario the search accepted 1 move in 55 attempts; the other 54 rejections were genuine (each proposed mission evaluated worse under exact evaluation), not search noise.

The verification battery

Nine checks run on any campaign set the system might publish. Quadrant purity verifies every campaign's layers share one quadrant. Tank count verifies the five-tank limit. Stack order reconstructs each tank's layer sequence and checks the mission-global in-flight semantics: for

adjacent layers, the deeper layer's first take may not precede the campaign in which the shallower layer's last take completes, unless they share that campaign. Precedence walks the southeast rule table (AP-101 before AP-102, AZ-101, AZ-102, AY-101; AZ-101 before AZ-102 and AY-101; AN-101 before AN-106; AW-104 before AW-103 and AW-105) and checks first takes against predecessor drain campaigns. The band check flags over-band volumes as violations and sub-band volumes as warnings unless the campaign is a legal final drain. Min-transfer checks the 38 kL floor with the final-drain exemption against running cumulative takes. Mass and waste-oxide conservation close per layer and per mission. WAC compliance evaluates every campaign blend on the slurry-basis convention of the export workbook against the true regulatory limits.

The sweep harness

The sweep driver enumerates the 32 published scenarios from the custom-scenario registry, orders them largest-first for load balance, and runs one subprocess per scenario with 10 concurrent workers on a 10-core machine. Each worker builds the baseline, runs the LNS, verifies the result, and writes a JSON report (campaigns with layer takes, exact evaluations, cache statistics, wall time, verification output). The aggregator reads all reports, recomputes the suite rollup, and writes the summary table. Failures are per-scenario and never contaminate neighbors; the first sweep run's per-scenario logs are what surfaced the builder defect described next.

Discussion

What the linearity result means for future work

The single most useful thing this study produced is not an optimizer but a bound. Inverse waste loading is linear in oxide composition to $R^2 = 0.79$ on 76 exact evaluations, and the residual is neither large nor predictable (a grouped quadratic on alkali, sulfate-halide, alumina, silica, and iron features recovers only 20% of its variance). Mission glass therefore has a composition floor that no partitioning scheme can beat by more than the waste-weighted residual spread, which the data places at roughly one percent. Teams should treat claims of large canister savings from blending rearrangement with that bound in mind, and should test any proposed blend change against exact evaluation before believing it. The conclusion drawn from this observation is that future attempts to optimize canister minimization should focus on segregation of high-level and low-level waste streams and conditioning of the waste. Otherwise, the composition of the tank farms waste serves as an optimization floor.

The bound also explains a result that initially looked like a failure. The search accepted one move on the reference scenario in 55 tries. It is not that the search failed to explore; the ruin operators proposed 54 alternative missions, every one scored exactly, and every one was worse or illegal. The baseline sits close to the achievable floor for this scenario. The same held across the suite: 3,970 exact evaluations of candidate repairs produced 31 accepted moves in total (out of 1,629 ruin-and-repair attempts).

De-mixing as a first-class candidate

The AN-106 case deserves a general statement. Waste loading is limited by the worst-constrained glass property of the blend. When two layers are individually hot on the same property (here, glass-target species rows), their blend is at least as hot, and blending buys nothing on that property while diluting every other constraint's headroom. The baseline's blends of AN-106_L1 with AP-102_L3 ran at waste loading 0.289 to 0.302 where the pure layers ran 0.30 to 0.52. Classical blending intuition, formed on cases where partners are complementary, does not transfer to cases where they are similar. The practical remedy is cheap: always generate the pure-split repair alongside the blended one and let exact evaluation pick. On this suite the arbitration between the two candidate classes was decided by margins as small as one canister, so neither class can be dropped.

The cost of retrieval-legal acceptance

The verification gate reduced suite-wide savings from 849 to 216 canisters. The gap is not search noise; it is the value of violating retrieval sequencing, and it should be understood before anyone is tempted to reclaim it. Most of the ungated savings came from repair splices that reordered campaigns freely, which in a real flowsheet means starting a layer before the one above it is drained, or feeding a successor tank before its predecessor. Some sequencing constraints might be relaxable with the subject matter experts (the in-flight boundary rule is a candidate, since its cost shows up repeatedly in the chain repacks), and any such relaxation would be a deliberate modeling decision with its own verification change, not a silent optimization gain. Until then, 216 canisters and 457 campaigns is the honest result.

The compliance gate raises a subtler point. The screening limits inside the repair LP are climbed to their true values before falling back to the paper's 60% scalings, which means the LP routinely proposes blends the paper's mixing allowance would refuse. The acceptance gate then checks those blends against exact glass-level evaluation, the basis on which the facility's reports and the regulatory limits actually operate. Under that posture the climbs are not relaxations at all; they

are removals of a conservative approximation that the exact evaluation renders moot. Teams with a different compliance posture, one that binds at the screening stage, should set the ladder's top rung to their binding basis and re-run; the architecture isolates that decision to one constant.

Limitations

The search is a hill climb, so its results depend on the ruin operators' ability to reach improving neighborhoods. The operator set was tuned on one scenario; other weightings might find moves on the 15 scenarios that did not improve, though the accepted-move rate (23 accepted across 3,970 exact evaluations) suggests those scenarios are genuinely at their floor rather than underexplored. The destroy-set sizes cap at the whole chain for the tank-chain operator; larger structural moves (repartitioning across quadrants) are excluded by the quadrant rule, which the mission's flowsheet fixes.

The objective counts canisters, not operations. Scenario 135.120.200 shows the consequence: 3 extra campaigns bought 51 canisters. If the team's utility function prices campaign overhead explicitly (sampling, paperwork, recipe changes), the acceptance gate can compare a weighted objective instead; the code isolates this to the acceptance comparison. The current paper reports both quantities per scenario so readers can apply their own weighting.

Exact evaluation inherits the EWG2.5 model's own limitations, documented in PNNL-35884 and in the facility modeling team's earlier white paper: per-model basis conventions, the Batch 45 alumina provenance exception, and the property-covariance treatment. Nothing in this work changes those; the optimizer consumes the same formulation solves the app has always reported.

Data provenance and reproducibility

All scenario data comes from the October 2026 Best-Basis Inventory retrieval and the Appendix D layer-by-layer vectors of RPP-RPT-64878, through the same ingestion path the production app uses, including homogenized heel layers for tanks outside the published set. Every headline number in this paper was re-derived from the cached JSON reports during writing: the suite rollup from lns_sweep_summary.json, the scenario detail from lns_report_135.120.150.json, the linearity fit from residual_model.json, and the baseline structure from mission_report_135.120.150.json. The optimizer, sweep harness, verification battery, and reports ship on the optimized-mission branch; the figures regenerate from the cached reports by one script.

Computational cost in practice

The full 32-scenario sweep ran in 7.6 core-hours, wall time about 50 minutes on a 10-core desktop, with the exact evaluation memo providing a 29-fold reduction in SLSQP calls. A single-scenario run costs 11 to 24 minutes. This is fast enough to run overnight after any change to inventory data, WAC limits, or the builder, which is the usage pattern we recommend: the optimizer is a check on the baseline, not a replacement for it, and its accepted moves (with their exact evaluations and verification reports) are meant to be reviewed campaign by campaign before adoption into a published mission plan.

Conclusions

A large-neighborhood search with exact, memoized glass evaluation and verification-gated acceptance improves the Section 6.2 campaign retrieval model across all 32 published scenarios without ever degrading a mission. The suite-wide result is 216 fewer canisters (0.057%) and 457 fewer campaigns (20.0%), with 17 scenarios improved, 15 unchanged, none worse, and all hard retrieval rules verified on every published result.

The study's structural finding is that canister count sits at a composition floor: inverse waste loading is nearly linear in blend composition, mission glass is nearly invariant to how waste is partitioned, and the from-scratch builders that ignored this lost to the baseline by construction. Optimization effort for this problem belongs in campaign structure, where the paper's own builder leaves artifact chains that a chain-aware search repairs cleanly, and the potential further savings from any composition-rearranging scheme are bounded near one percent.

Three practices are recommended. First, run the verification-gated LNS as a post-processing check on any published mission plan; it is cheap, it can only improve, and its per-move reports make every change auditable. Second, gate blend changes on exact evaluation rather than screening-level scores; the suite's 29-to-1 cache hit ratio makes this practical at scale. Third, when proposing blends, generate pure-run alternatives alongside them; the assumption that blending helps is wrong often enough that arbitration is required, not optional.

The campaign builder defect documented here, in which the refinement LP read the most recently built scenario's data, is fixed on the same branch and should be merged independently of the optimizer decision, since every multi-scenario caller of the production entry point hits it.

References

RPP-RPT-64878 Rev 1, Direct-Feed High-Level Waste Feed Vectors Assessment, Washington River Protection Solutions, Richland, WA (2021). Section 6.2 campaign generation; Appendix D layer-by-layer feed vectors; Table A-30 mission anchors.

PNNL-35884 Rev 1, Equations and Functions for Estimating Waste Glass Properties as Part of HLW Glass Formulation, Pacific Northwest National Laboratory, Richland, WA. EWG2.5 property and composition models.

Pisinger, D., and Ropke, S., "A General Heuristic for Vehicle Routing Problems," Computers & Operations Research 34(8), 2403-2435 (2007). Ruin-and-recreate (large neighborhood search) framework.

Ropke, S., and Pisinger, D., "An Adaptive Large Neighborhood Search Heuristic for the Pickup and Delivery Problem with Time Windows," Transportation Science 40(4), 455-472 (2006). Adaptive operator weighting; acceptance principles.

Shuford, D. M., and Stegen, G. R., River Protection Project Mission Analysis: Waste Blending Study (SRNL-STI-2010-00557), Savannah River National Laboratory, Aiken, SC (2010). Terminal mission glass minimization via blending.

Kim, D.-S., et al., HLW Glass Groupings for Waste Loading Optimization (ORNL/TM-2011/53), Oak Ridge National Laboratory, Oak Ridge, TN (2011). Limiting-factor taxonomy for HLW blend grouping.

Hoza, M. M., Multipurpose Optimization Models for the HLW Waste Blend System (WSRC-RP-94-1017), Westinghouse Savannah River Company, Aiken, SC (1994). OWL blending optimization.

HLW System Modeling Team, Coupled Mission Planning and Facility Process Modeling for Hanford High-Level Waste Vitrification, internal white paper, Tank Waste Operations Modeling (October 2026). Prior architecture paper describing the EWG campaign builder and the coupled facility model.

Best-Basis Inventory data retrieval via the PNNL PHOENIX data service, October 2026 snapshot; ingestion and layer-vector construction per the data-lineage section of the prior white paper.